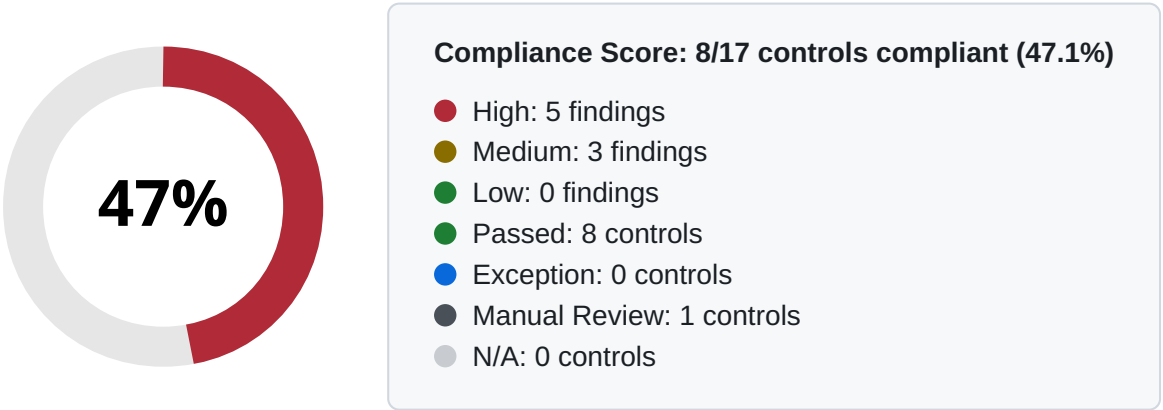


Cisco IOS-XE Compliance Report

CONFIDENTIAL - INTERNAL USE

Device:	device_config
Device role:	edge-router
Audit date:	2026-08-26
controls.yaml version:	VibeCoding controls.yaml v6f92236
Generated by:	VibeCoding Compliance Report Generator

Executive Summary



METRIC	COUNT
Total controls evaluated	17
Compliant	8
Non-compliant (High)	5
Non-compliant (Medium)	3
Non-compliant (Low)	0
Exception granted	0
Manual review required	1
Not applicable	0

Scope & Methodology

This audit evaluated the device configuration listed above against the VibeCoding controls framework (controls.yaml v6f92236), covering the VibeCoding 3-plane hardening model (management, control, and data plane) and the Cisco IOS-XE hardening practices it

encodes. Controls without a mapped external CIS Cisco IOS-XE Benchmark section are cited by their VibeCoding control ID only; no CIS section reference is claimed unless a real mapping has been supplied for that control.

Deterministic controls were evaluated automatically against the device's running configuration. Controls marked "Manual Review" require human judgment (e.g. banner wording against corporate legal policy) and are listed separately below rather than auto-scored. Findings already covered by a documented exception are still listed in Findings Detail, marked accordingly, rather than silently excluded.

Out of scope: physical security, change-management process compliance, and any control outside the VibeCoding framework's current 17 controls.

Findings Detail

[control_00004] TACACS — HIGH

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

Failure to communicate with central security servers forces the use of local accounts, making user management difficult and less secure across the network.

Evidence

'tacacs server ACME-TEAM-TACACS-01' uses a reversible Type 7 key; Type 6 encryption ('key config-key password-encrypt' + 'key 6 <key>') is required.; 'tacacs server ACME-TEAM-TACACS-02' uses a reversible Type 7 key; Type 6 encryption ('key config-key password-encrypt' + 'key 6 <key>') is required.; TACACS group name 'ACME-TEAM-TACACS' does not match the corporate group 'ACME_TACACS_SERVER_GROUP'.; No 'ip tacacs source-interface' configured under the TACACS server group.

Remediation Command

```
aaa new-model
! Repeat the following 'tacacs server' block for each TACACS+
server.
tacacs server <tacacs_server_name>
  address ipv4 <tacacs_server_address>
  key 6 <tacacs_key>
  timeout <tacacs_timeout>
aaa group server tacacs+ <tacacs_group_name>
  server name <tacacs_server_name>
  ip tacacs source-interface <source_interface>
```

Operator Inputs Required

<source_interface>, <tacacs_group_name>, <tacacs_key>,
<tacacs_server_address>, <tacacs_server_name>, <tacacs_timeout>

Owner

Network Security Team

Target Remediation Date

2026-09-10 (<= 15 days, or next maintenance window)

[control_00005] Local and emergency users — HIGH

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

Lack of local emergency accounts can result in total loss of access during a network or TACACS outage.

Evidence

Local account(s) use a weak password-hashing type: ['5'] (expected type 8/9 scrypt).;
Local accounts use inconsistent password-hashing types: ['5', '8'].

Remediation Command

```
username <user> privilege <privilege_level> algorithm-type  
scrypt secret <password>  
enable secret <secret_psswd>
```

Operator Inputs Required

<password>, <privilege_level>, <secret_psswd>, <user>

Owner

Network Security Team

Target Remediation Date

2026-09-10 (<= 15 days, or next maintenance window)

[control_00011] SNMP — HIGH

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

SNMPv1/v2c transmit community strings in clear text.

Evidence

SNMPv3 group with 'priv' is not configured.; SNMPv3 user is not created with SHA/AES configuration.; SNMP trap host is missing.; 'snmp-server enable traps' command is missing.

Remediation Command

```
snmp-server group <group_name> v3 priv
snmp-server user <user_name> <group_name> v3 auth sha
<auth_password> priv aes 128 <priv_password>
snmp-server host <snmp_server_ip> version 3 priv <user_name>
snmp-server enable traps
```

Operator Inputs Required

<auth_password>, <group_name>, <priv_password>, <snmp_server_ip>, <user_name>

Owner

Network Security Team

Target Remediation Date

2026-09-10 (<= 15 days, or next maintenance window)

[control_00014] VTY Lines — HIGH

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

Open VTY/Console sessions without timeouts can be hijacked.

Evidence

'line con 0': 'login local' is missing.; 'line con 0': 'exec-timeout' is missing.; 'line vty 0 4': 'login local' is missing.; ACL 'VTY-OPEN' bound to VTY lines is permissive instead of restrictive: permit tcp any any eq 22; permit tcp any any eq telnet; 'line vty 5 15': 'login local' is missing.

Remediation Command

```
line con 0
  password <console_password>
  login local
  exec-timeout <minutes> <seconds>
  logging synchronous
  exit
line vty <start_line_number> <end_line_number>
  access-class <vtty_acl_name> in
  password <vtty_password>
  login local
  exec-timeout <minutes> <seconds>
  transport input <allowed_protocols>
  logging synchronous
  exit
```

Operator Inputs Required

<allowed_protocols>, <console_password>, <end_line_number>, <minutes>, <seconds>, <start_line_number>, <vtty_acl_name>, <vtty_password>

Owner

Network Security Team

Target Remediation Date

2026-09-10 (<= 15 days, or next maintenance window)

[control_00016] Mandatory Commands — HIGH

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

Omitting baseline security commands, including CoPP, leaves the device vulnerable to common automated attacks, misconfigurations, and control-plane resource exhaustion (CPU/DoS) from unfiltered traffic.

Evidence

No 'service-policy input <policy_name>' applied under 'control-plane' (missing Control Plane Policing).

Remediation Command

```
control-plane
service-policy input <copp_policy_name>
```

Operator Inputs Required

<copp_policy_name>

Owner

Network Security Team

Target Remediation Date

2026-09-10 (<= 15 days, or next maintenance window)

[control_00009] NTP — MEDIUM

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

Drifting clocks lead to inaccurate log timestamps, making forensic analysis and event correlation during a security incident nearly impossible.

Evidence

'prefer' keyword is missing on the primary NTP server.; NTP authentication is not configured ('ntp authenticate' / 'ntp authentication-key' / 'ntp trusted-key' missing).

Remediation Command

```
ntp authentication-key <ntp_key_id> md5 <ntp_key_value>
ntp trusted-key <ntp_key_id>
ntp authenticate
ntp server <ntp_ip_address> prefer
ntp server <secondary_ntp_ip_address>
```

Operator Inputs Required

<ntp_ip_address>, <ntp_key_id>, <ntp_key_value>,
<secondary_ntp_ip_address>

Owner

Network Security Team

Target Remediation Date

2026-09-25 (<= 30 days)

[control_00010] Syslog — MEDIUM

Status

NON-COMPLIANT

Reference

VibeCoding controls.yaml v6f92236

Risk Statement

Without centralized logging, local logs may be overwritten or cleared by an attacker, leaving no trace of unauthorized activity.

Evidence

'logging on' command is missing.; Logging trap level is not set to 'informational'.; 'logging source-interface' is not configured.

Remediation Command

```
logging on
logging host <ip_syslog_server>
logging trap informational
logging source-interface <syslog_interface>
```

Operator Inputs Required

<ip_syslog_server>, <syslog_interface>

Owner

Network Security Team

Target Remediation Date

2026-09-25 (<= 30 days)

[control_00017] Recommended Commands — MEDIUM

Status
NON-COMPLIANT

Reference
VibeCoding controls.yaml v6f92236

Risk Statement
Without brute-force protection, attackers can use automated scripts to guess local passwords without being blocked or logged.

Evidence
'login delay' is missing.; Scrypt algorithm (type 9) is not configured for any local user account.

Remediation Command

```
username <local_admin> privilege 15 algorithm-type scrypt
secret <strong_password>
login block-for <blocked_time> attempts <attempts_number>
within <attempts_time>
login delay <login_delay>
login on-failure log
login on-success log
security passwords min-length <password_length>
```

Operator Inputs Required
<attempts_number>, <attempts_time>, <blocked_time>, <local_admin>, <login_delay>, <password_length>, <strong_password>

Owner
Network Security Team

Target Remediation Date
2026-09-25 (<= 30 days)

Manual Review Required

The following controls cannot be automatically scored and require human review against corporate policy.

CONTROL	TITLE	EVIDENCE
control_00012	Banners	'banner motd' section in the running configuration.

Full Control Appendix

All 17 evaluated controls, regardless of outcome.

CONTROL	TITLE	SEVERITY	STATUS
control_00001	Hostname	Low	Compliant
control_00002	Domain	Low	Compliant
control_00003	AAA	High	Compliant
control_00004	TACACS	High	Non-Compliant
control_00005	Local and emergency users	High	Non-Compliant
control_00006	SSH	High	Compliant
control_00008	Telnet Blocking	High	Compliant
control_00009	NTP	Medium	Non-Compliant
control_00010	Syslog	Medium	Non-Compliant
control_00011	SNMP	High	Non-Compliant
control_00012	Banners	Medium	Manual Review
control_00013	Passwords	High	Compliant
control_00014	VTY Lines	High	Non-Compliant
control_00015	Basic Security	High	Compliant
control_00016	Mandatory Commands	High	Non-Compliant
control_00017	Recommended Commands	Medium	Non-Compliant
control_00018	Prohibited Commands	High	Compliant

Remediation Roadmap

SEVERITY	CONTROL	OWNER	TARGET REMEDIATION DATE
High	[control_00004] TACACS	Network Security Team	2026-09-10 (<= 15 days, or next maintenance window)
High	[control_00005] Local and emergency users	Network Security Team	2026-09-10 (<= 15 days, or next maintenance window)
High	[control_00011] SNMP	Network Security Team	2026-09-10 (<= 15 days, or next maintenance window)
High	[control_00014] VTY Lines		2026-09-10 (<= 15 days, or next maintenance window)

SEVERITY	CONTROL	OWNER	TARGET REMEDIATION DATE
		Network Security Team	
High	[control_00016] Mandatory Commands	Network Security Team	2026-09-10 (<= 15 days, or next maintenance window)
Medium	[control_00009] NTP	Network Security Team	2026-09-25 (<= 30 days)
Medium	[control_00010] Syslog	Network Security Team	2026-09-25 (<= 30 days)
Medium	[control_00017] Recommended Commands	Network Security Team	2026-09-25 (<= 30 days)

Sign-off

Network Team - Name / Signature	Role	Date
Security Team - Name / Signature	Role	Date